Formalisation of the problem: chromosome and fitness function

Navigation graph and fitness function

G = (V, E): V – water points (OPEN_SEA, COASTAL_SEA) of the area;

E – k nearest neighbours closer than l_max, segments that cross land are removed (visibility test).

Edge cost: $c(e) = l(e) \cdot (1 + \alpha \cdot s(e) + \beta \cdot \varphi(e))$, $\varphi(e) = \max(0, 1 - d_land(e) / r)$

Fitness: $F(P) = \Sigma\, c(e) = L(P) + \alpha \cdot Lc(P) + \beta \cdot Ld(P) \rightarrow \min$

Quality: $\delta = (F_GA - F^*) / F^* \cdot 100\ \%$, F^* – optimum of Dijkstra / A*

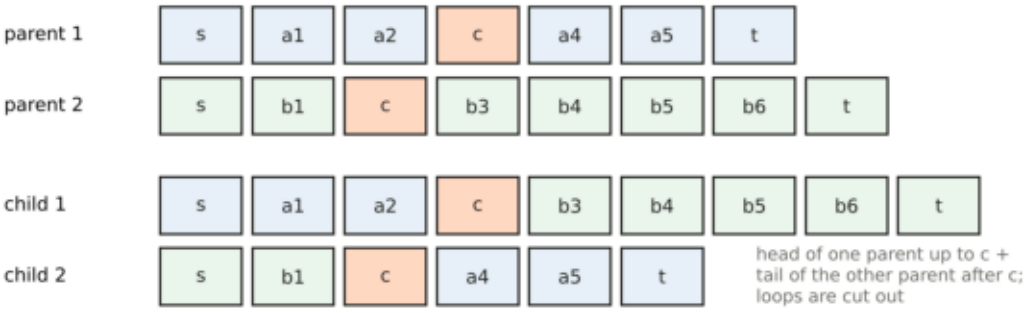
l(e) – great-circle length (haversine), s(e) – share in coastal water, d_land(e) – distance to land,

α, β – penalties, r – radius of the danger zone. $c(e) \geq l(e)$, so $h = d_gc$ is admissible for A*.

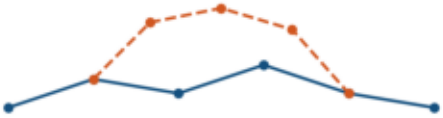
a) chromosome = route of variable length

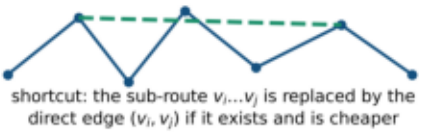


b) one-point crossover at a common node c



c) mutations





Initial population: random walks s → t, $p(v|u) \sim \exp(\kappa \cdot \gamma)$, $\kappa \in [1; 15]$, 50 % through an intermediate node

Selection: tournament (k = 3); crossover: one point at a common node, $p_c = 0.8$

Mutation ($p_m = 0.3$): re-routing (60 %) or shortcut (40 %); elitism: e = 2

Stop: $G_{max} = 300$ generations, 80 generations without improvement or user request

Rev.	Sheet	Doc. No.	Sign.	Date
Developed		Li H.		
Checked		Kozinsky A.A.		
T. contr.				
N. contr.				
Approved		Golovko V.A.		

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Development of an application for finding the optimal course of a sea vessel over open-sea points based on a genetic algorithm

Formalisation (poster)

Lit.	Mass	Scale
	–	–

Sheet 3 | Sheets 5

BrSTU, Dept. of IIT
group ИИ-2025